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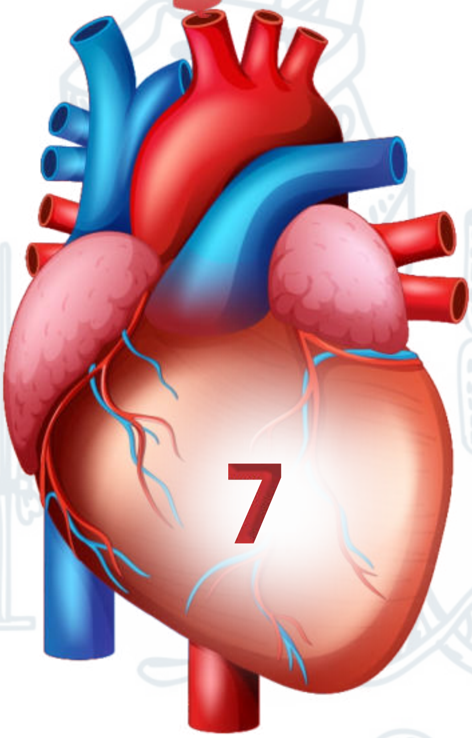

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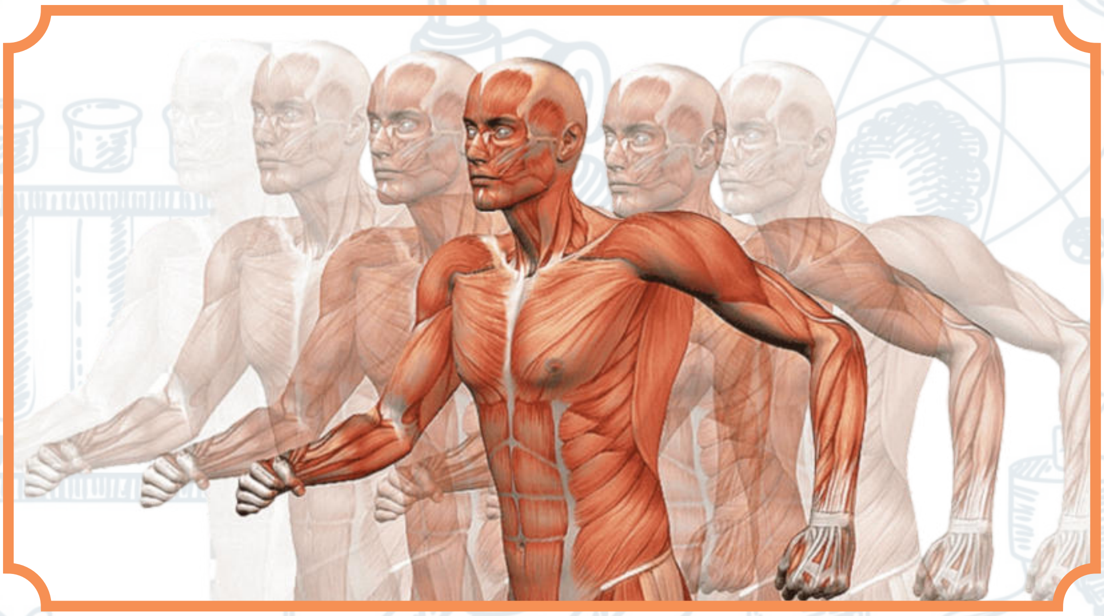
PROTOCTISTS & FUNGI



KINGDOM PROTOCTISTA:

The Kingdom Protista is a group of organism that are eukaryotic ,but are instead eukaryotic. Protists are a Diverse group that can be found in many different environments, including ocean, puddles, and even, b/w soil particles. They are can be single celled or Multicellular,and can be heterotrophic or autotrophic. Example: Amoebas, Algae, Diatorm as well Plasmodium and Tryanosoma, the parasites which cause malaria & sleeping sickness recpectively.

THE EVOLUTIONARY RELATIONSHIP:

The evolutionary relationship of Protists have molecular Phylogenetics, the sequencing of entire genomes, and Transcriptomes, and electron microscopy studies of the flagella apparatus & cytoskeleton. Therefore biologists regard (Propetista) kingdom as a Polyphletic group of organism due to Diversification.

MAJOR GROUP OF PROTOCTISTA:

The Protoctista includes three group which formerly had separate taxonomic status.

i. Plants like Protoctista-------Algae
ii. Fungi like Protoctista-------Primitive Fungi
iii. Animals like Protoctista------------ protozoa

PLANTS LIKE PROTOCTISTA-------ALGAE:

Plants like protista that perfrom photosynthesis the process by autotrophs convert sunlight into a usable from of energy

→ Have no specialized tissue / organ.
→ Are classified based on difference in structure & type of pigments that they contain.
----- →Can be unicellular or multicellular.

CHARACTERSTIC OF ALGAE:

→Algae contain chlorophyll in their chloroplasts to produce food & O2 through photosynthesis
→The cell wall is present and have composition like plants cell wall
→Their cytoplasm usually contain one or more large vacuoles.

ULVA :

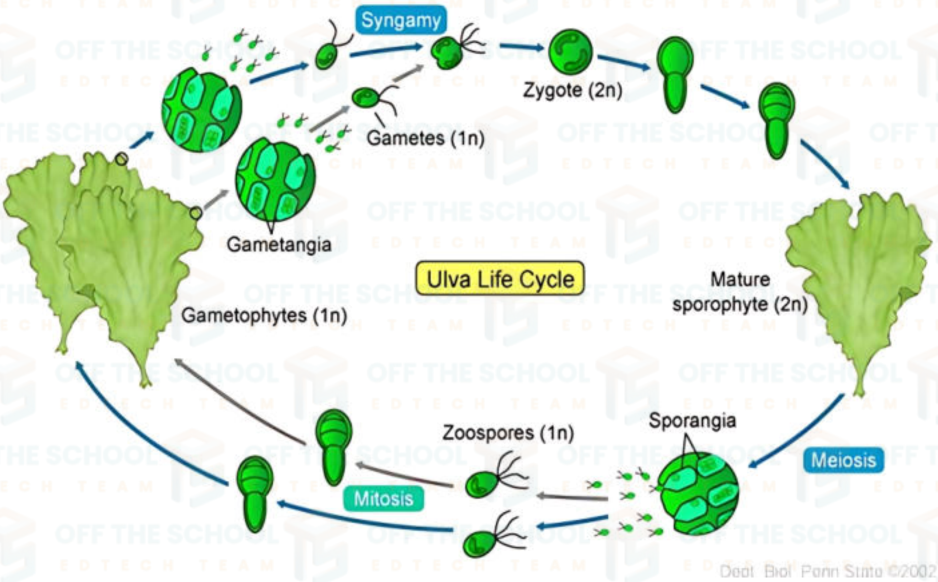
Ulva is essentially a marina aldae generally found in rocky shores when its occurs attached to stones, rocks, its usually grow in association with various other algal species such as cladophora. Thallus in ulvas is of two types. The one with 26 chromosomes is called Sporophyte . Other with 13 chromosomes called Gametophyte. Morphologically both gametophyte and sporophyte exactly alike hence called Isomorphic and also reproduces asexually a well as sexually, alternating

ASEXUAL REPRODUCTION :

Asexual reproduction take place in asexual Ulva(Diploid sporophyte with 26 chormosomes or 2m) there spores are formed by meiosis in all cellof the body 8 to 16 haploid zoopores are formed ina singal parent cell. The liberated haploid zoospores after a period of swimming & rest their flagella and grow into gametophyte ulva thalli.

SEXUAL REPRODUCTION:

Sexual reproduction is isogamous the type of sexual reproduction takes place by the fusin of morphologically & physiologoically gametes.they produced in sexual plants (haploid gametophytes with 13 chromosomes
(n). ulva thallus (sporophyte) which is similar to sexual thallus (Gametophyte) in morophology.



ALTERNATING OF GENERATION IN ULVA:

The type of life cycle asexual reproduction alternation with sexual reproduction. Which are similar in morphology but differ in choromosome number.

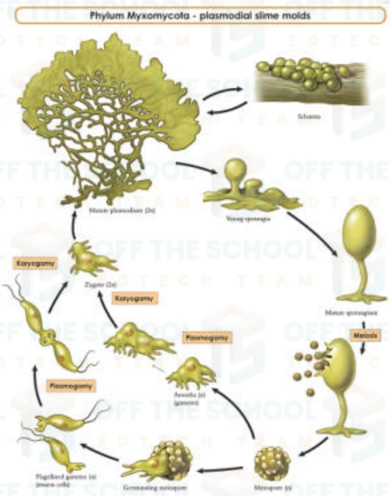
FUNGI LIKE PROTOCTISTA:

These organism superficially remeble to fungi. The body called MYCELIUM. They are hetertrophic absorptive feeder. The member of non motile are few develop movement at some stages in their live Major group of fungi like protoctista are slime molds & water molds.

SLIIME MOLD:(MYXOMYCOTA)

[Motile like animals & produce spores like fungi]

slime mold are amoeboid protoctists that produced fruiting bodies. They were often classified with fungi because they absorb nutrient directly from the environment the most conspicuous part of life the cycle which is a small slimy mass. A smile mold is a strange & truly wonderful thing, Its structure behavior .some species of slime molds are creeping masses of living substance. Plasmodium move along the forest floor vacuoles, & dead leaves absorebed sunlight. Part of each sporangium called capsule produce a large no of microscope, asexual reproduction cell called spores.we would call the organism a plants.



WATER MOLDS (OMYCETES):

The Oomycetes, or Oomycota from a distinct phylogenetic lineage of fungus-like eukaryotic microorganism within the stramenopiles. Thet are filamentous and heterotrophic, & can reproduce both sexually & Asexually.

PHYTOPHTHORA INFESTANS:

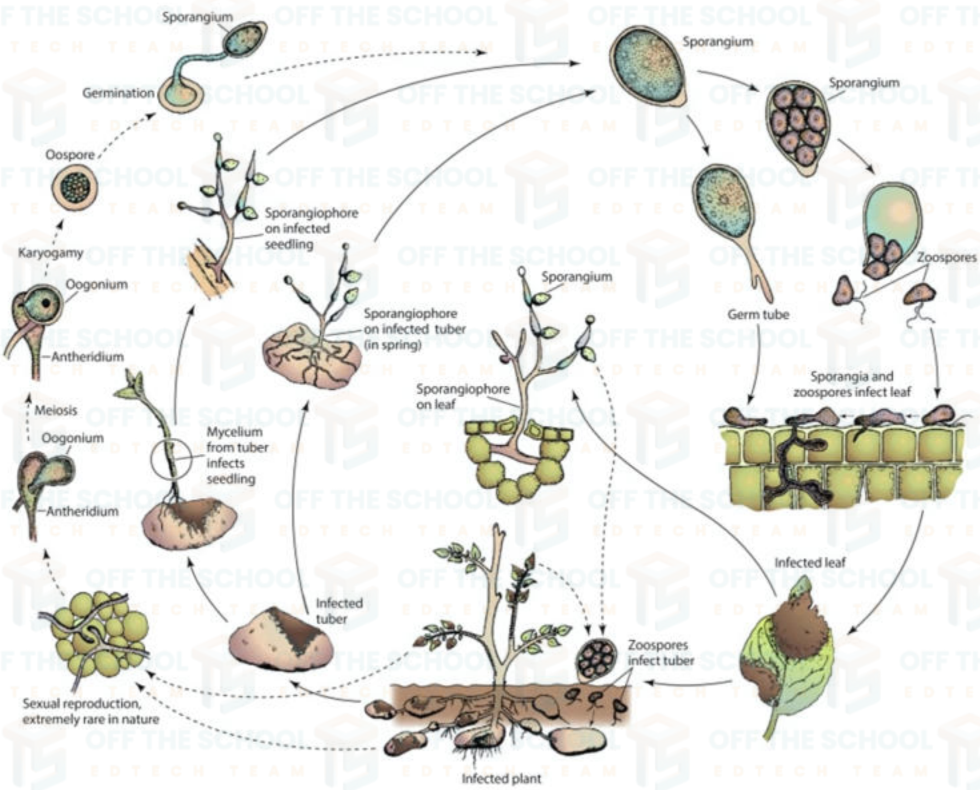
The fungus is responsible for tomato late blight .it us feared by tomato potato producer its effect all types of production more intensive or extensive.

STRUCTURE OF MYCELIUM:

The mycelium is a porous, branching structure of tabular filaments called hyphae that make up the body of a fungus.

REPRODUCTION:

Reproduction take place by mean of sexual & asexual method. Asexual reproduction is Oogamous or sexual reproduction is mean is biflagellated. The female sex organ is Oogonium & male organ is antheridium. The fertilized oospore germinates to from a sporangium at the tip of germ tube.



ANIMALS LIKE PROTOSTISTA -----PROTOZOA:
Protozoans from a large group of protoctista .they are unicellular animals like organism with ingestive heterotrophic nutrition.they found in water presents.free living are various method of locomotion.
Members of this group of protozoa are divided in five classes and their mean is locomotion.
→class sarcodina(rhizopoda)
→ class flagellate(mastigophora)
→class ciliate(ciliophores)
→class suctoria → class sporozoa

IMPORTANCE OF PROTOCTISTA:
Protistans have great biological importance & significance to human.

PATHOGENIC PROTOCTISTA:
Pathogenic protoctista are unicellular parasites that infect other organism & cause disease.
Trypanosoma, Plasmodium, Toxoplasma, Entamoeba, Histolytica, cryptosporidium. Pathogenic protista enter the human host to feed, grow & reproduce they are harm full host.

FERTILLIZER
A fertilizer is any material of natural or synthetic origin that is applied to soil source of NPK in various region of the world.

ALTERNATION SOURES OF FOOD:
Chlorella is considered alternate source of energy. Red and brown algae are a potential source of food

BOITECHNOLOGY:
Algae are important in biotechnology because they are versetile & sustainable organism use for food, energy medicines, e.t.c. fuel deficiency will become great in future. Botrycoccus brauniiis a green pyramid-shaped planktonic microalga.

FUNGI:
The study of fungi, group that includes the mushroom & yeast. Many fungi are useful in medicine & industry. Mycological research has led to the development of such antibiotic drugs as penicillin, streptomycin, and tetracycline, as well as other drugs the study of fungi is called mycology

GENERAL CHARACTERSTICS:
Fungi are multicellular (expect yeast) eukaryotic, non chlorophyllous, absorptive hetetrophs. They may be saprotrophs, parasites, mutulist and predator.

SAPROTROPHS:
→ Decompose cellulose and lignin.
PARASITES:
→ Facultative can grow on their host as well as by themselves on artificial growth media

PREDATORS:
→Oyster mushroom →nematodes.

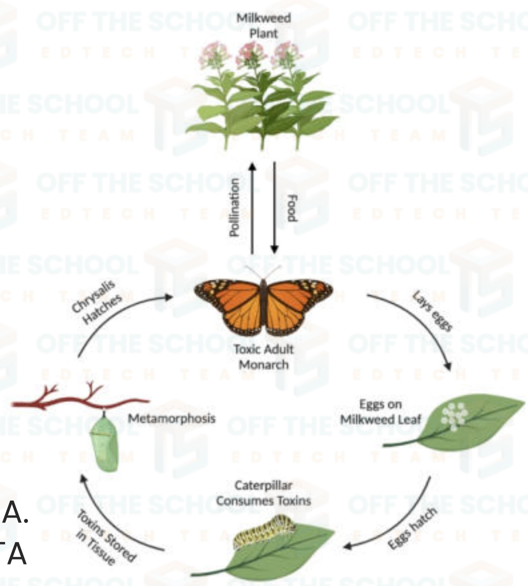
MUTUALISM:
Two key mutualistic symbiotic association.
The mycelium is the vegetative structure of fungi. Mycelia are made up fungi. Mycelia are made up of hyphae. Which are the individual branching cells of the fungi, sectioned by septa & surrounded by a strong cell wall made of chitin as the fibrillar material. Hyphae are the feathery filaments that make up multicellular fungi. Basidio spore are generally characterized by an attachment peg (called a hilar appendage) on its surface. Asco spore subsequent upon meiosis in zygote.

Diploid zygote nuclei form during sexual reproduction. They have a characterstic mitosis called nuclear mitosis During which nuclear membrance does not break & spindle is formed within nuclei.

Most fungi can reproduce asexually as well as sexually (except imperfect fungi where sexual reproduction has not been observed)

Fungi can reproduce asexually in a number of ways. Fragmentation a fungal mycelium can separated into pieces & each piece can grow into a new mycelium. Budding a bulge form on the side of a yeast cell, the nucleus divides, & the bud separates from the parent cell. Spore production fungi produce haploid spore they are genetically identical to the parent cell.

Fungi can also reproduce sexually, but most species of fungi reproduce asexually most of the time. Fungi have a haploid-dominant life cycle, this allow fungi to reproduce both asexually & sexually.



DIVERSITY AMONG FUNGI:
The diversity of fungi is astounding.
Described species of fungi.Fungi are classified into four groups:
→DIVISION ZYGOMYCOTA. → DIVISION BASIDIOMYCOTA.
→DIVISON ASCOMYCOTA → DIVISION DEUTEROMYCOTA

DIVISION ZYGOMYCOTA: are a relatively small group in the fungi kingdom

→Most zygomycota are saprobes, while a few species are parasites.

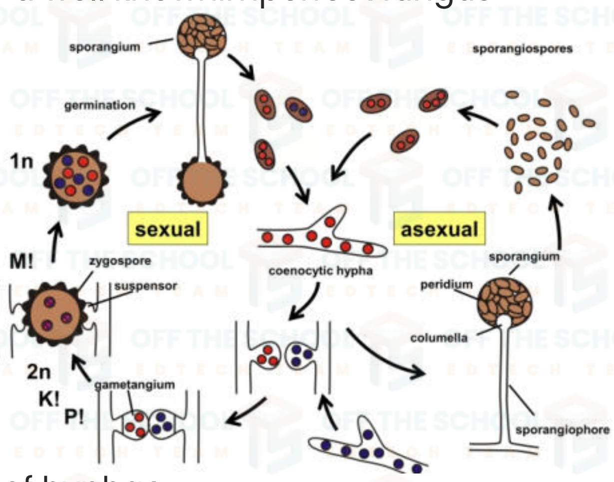
→Zygomycota usually reproduce asexually by producing sporangiospores & fragmentation.
→No fruiting body
→The result diploid zygospores remain dormant & protected by thick coats until environmental condition have improved.

DIVISION BASIDIOMYCOTA:
→Basidiomycota reproduce sexually through the formation of basidiospore. Basidiospore is a club-shaped organ.
→basidiomycota can have a wide range of appearance, including mushroom.
→Basidiomycota play a crucial role in decomposing organic matter & forming beneficial relationship with plants.
→Basidiomycota are filamentous fungi composed of hyphae, expect for basidiomycota-yeast, which is unicellular .with during mitosis.

DIVISION ASCOMYCOTA:
→Acomycota are septate fungi theie filaments are divided by cellular cross-wallcalled septa. Branch of hyphae have simple septal pores, that allow cytoplasm to move B/W cells.
→They act as plugs to prevent cytoplasm loss if a hyphal segment isinjured.
→which are sexual spore produced by Ascomycota.the are usually found in fruiting body called an ascocarp.
→Common examples of ascomycetes included yeast, powdery mildews

DIVISION DEUTEROMYCOTA:
→Deuteromycota are also known as anamorphic fungi or mitosporic fungi.
→they produce asexually usually by producing conidiospores.
→they are classified as fungi because they are multicellular tissue & erect hyphae with asexual spore.
→Penicillium the fungus that produce penicillin is a well knownimperfect fungus

LIFE CYCLE OF MUCOR (ZYGOMYCOTA):
Mucor is a genus of mold that is found in many places, including soil, plant debris & decaying fruit & vegetables. Mucor is erect hyphal sporangiophores form.



MYCELIUM:
Mycelium of mucor consist of long aseptale coenocytic hyphae.

TYPES OF HYPHAE: the mycelium consist of three of hyphae.

RHIZOIDAL HAPHAE:
Hyphae penetrate into the tissue of the substratum . they serve asthe fixtive & absorptive which are essential for nourishment.

EATING & DRINKING:
Mucor is a genus of fungi that can be found in many foods & is used in the preparation of fermented foods. mucor can cause food to spoil such as moldiness on bacon, egg pickle softening. Mucor can grow on the surface of cheee causing of defeats. The fermention process improve the flavor, texture, acidity, digestibility & shelf life of the food.

FOOD SPOILAGE:
Saprophytic fungi cause tremendous amount of spoilage of food stuff. Many toadstools contain poisonous alkaloid the effect the human nervous system.

PATHOGENIC FUNGI:
A number of important human and animals disease are caused by pathogenic fungal disease. The cause of the skin such as Ringworm. Athlete’s foot also an skin diease .fungal spore may trigger allgeric response in sensitive people.

HISTOPLASMOYTYSIS is a lung infection caused by inhaling fungal spores from the environment.

CANDIDA ALBICANS: is a yeast that can cause a fungal infection called candidiasis. They lives in your body located in your mouth, skin intestine. Many toad stool contain poisionous alkaloid that affect the human nervous system.

PLANTS DISEASE: is an abnormal condition that affect plants appearance or function, & can be caused by pathogenic or environmental factor. Non infectious or abiotic agents can also cause abnormal functioning in plants.

DECOMPOSER: Fungi are saprotrophic decomposer mean break down dead & decaying organic matter into simpler form.

SPOILAGE: Many fungi spoil leather, wool, books, timber, cotton e.t.c

ANTIBIOTICS: Are medication that treat bacterial infection by killing bacteria or preventing them from spreading. like; diarrhea ,pneumonia..e. t. c

BIOLOGICAL CONTROL: Are as bio control is a method of usingnatural organism to control pests, weeds, or pathogens.

FUNGI BIOTECHNOLOGY: Is a field of biotechnology that uses fungi to address environmental & societal issues.

GENETICS RESEARCH: Is a scientific study of genes, DNA & heredity in living organism to understands how traits are passed down from parents to offspring.

FUNGAL MUTUALISM: Is a symbiotic relationship where a fungus another organism benefit from each other.

MYCORHIZAE: Is a symbiotic relationship B/W plants & roots fungi that is beneficial to both the plants & the fungus

LICHENS: Is a complex organism that arte a symbiotic partnership B/W a fungus & or more algae or cyanobacteria.